

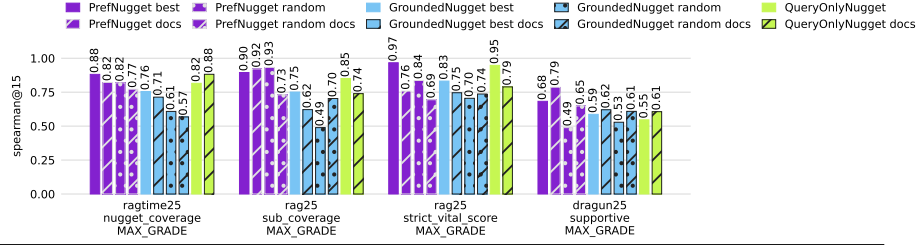
Too Many Questions: Full Empirical Result Analysis

1 Max Grade across Different Correlation Meta-Measures

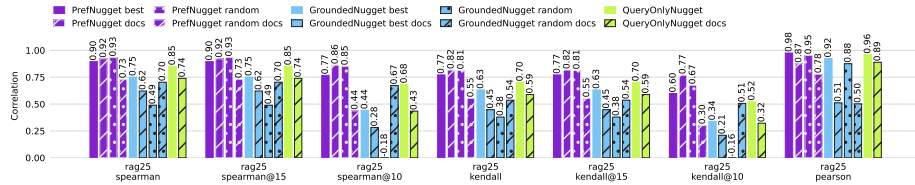
Correlation=spearman@15

Judge	graded	nugget	preference	ragtime25	rag25	rag25	dragun25
				nugget_coverage MAX_GRADE	sub_coverage MAX_GRADE	strict_vital_score MAX_GRADE	supportive MAX_GRADE
PrefNugget best	response	pairwise	best	0.8813	0.8950	0.9669	0.6828
PrefNugget docs	docs	pairwise	best	0.8219	0.9212	0.7580	0.7867
PrefNugget random	response	pairwise	random	0.8184	0.9326	0.8363	0.4888
PrefNugget random docs	docs	pairwise	random	0.7717	0.7303	0.6943	0.6486
GroundedNugget best	response	grounded	best	0.7552	0.7511	0.8318	0.5869
GroundedNugget best docs	docs	grounded	best	0.7135	0.6216	0.7459	0.6205
GroundedNugget random	response	grounded	random	0.6084	0.4889	0.7050	0.5286
GroundedNugget random docs	docs	grounded	random	0.5675	0.7035	0.7366	0.6089
QueryOnlyNugget	response	query	na	0.8155	0.8512	0.9477	0.5464
QueryOnlyNugget docs	docs	query	na	0.8818	0.7393	0.7890	0.6064

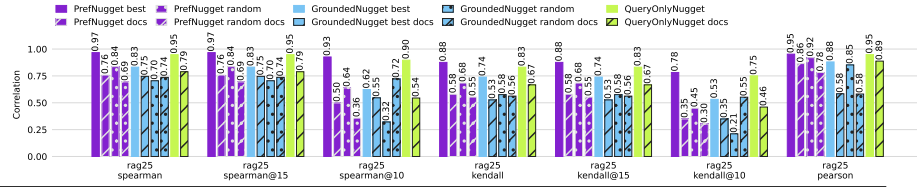
Category Wilcoxon Test		Category	Comparison	Metric	N	median_diff	p-value	winner	sig
		graded	docs vs response	spearman@15	20	-0.0412	0.4091	response	
		nugget	grounded vs pairwise	spearman@15	16	-0.1287	0.0010	pairwise	*
		preference	best vs random	spearman@15	16	+0.0952	0.0017	best	*



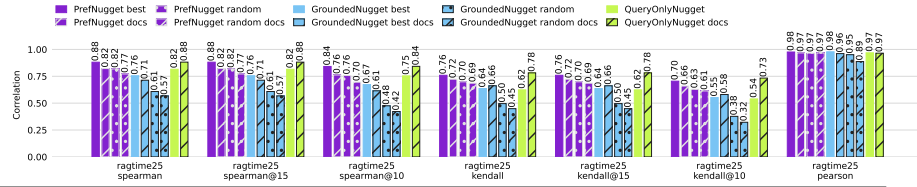
Judge	graded	nugget	preference	rag25					
				sub_coverage		MAX_GRADE			
				spearman	spearman@15	spearman@10	kendall	kendall@15	kendall@10
				pearson					
PrefNugget best	response	pairwise	best	0.8950	0.8950	0.7669	0.7714	0.7714	0.5981
PrefNugget docs	docs	pairwise	best	0.9212	0.9212	0.8616	0.8161	0.8161	0.7683
PrefNugget random	response	pairwise	random	0.9326	0.9326	0.8511	0.8129	0.8129	0.6742
PrefNugget random docs	docs	pairwise	random	0.7303	0.7303	0.4370	0.5528	0.5528	0.3027
GroundedNugget best	response	grounded	best	0.7511	0.7511	0.4425	0.6316	0.6316	0.3386
GroundedNugget best docs	docs	grounded	best	0.6216	0.6216	0.2822	0.4475	0.4475	0.2095
GroundedNugget random	response	grounded	random	0.4889	0.4889	-0.1770	0.3792	0.3792	-0.1581
GroundedNugget random docs	docs	grounded	random	0.7035	0.7035	0.6728	0.5361	0.5361	0.5060
QueryOnlyNugget	response	query	na	0.8512	0.8512	0.6829	0.7014	0.7014	0.5229
QueryOnlyNugget docs	docs	query	na	0.7393	0.7393	0.4343	0.5883	0.5883	0.3220



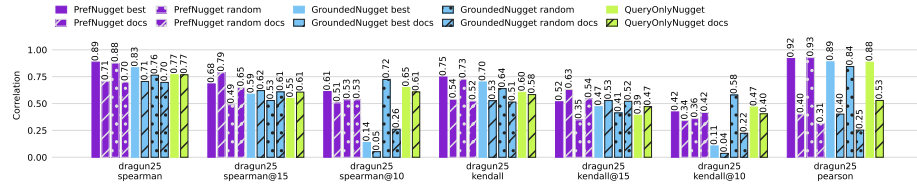
rag25										
strict_vital_score										
MAX_GRADE										
Judge	graded	nugget	preference	spearman	spearman@15	spearman@10	kendall	kendall@15	kendall@10	pearson
PrefNugget best	response	pairwise	best	0.9669	0.9669	0.9265	0.8760	0.8760	0.7821	0.9528
PrefNugget docs	docs	pairwise	best	0.7580	0.7580	0.4985	0.5791	0.5791	0.3492	0.8648
PrefNugget random	response	pairwise	random	0.8363	0.8363	0.6383	0.6839	0.6839	0.4495	0.9197
PrefNugget random docs	docs	pairwise	random	0.6943	0.6943	0.3570	0.5528	0.5528	0.3027	0.7813
GroundedNugget best	response	grounded	best	0.8318	0.8318	0.6233	0.7391	0.7391	0.5320	0.8802
GroundedNugget best docs	docs	grounded	best	0.7459	0.7459	0.5461	0.5265	0.5265	0.3492	0.5837
GroundedNugget random	response	grounded	random	0.7050	0.7050	0.3213	0.5758	0.5758	0.2108	0.8521
GroundedNugget random docs	docs	grounded	random	0.7366	0.7366	0.7217	0.5622	0.5622	0.5521	0.5816
QueryOnlyNugget	response	query	na	0.9477	0.9477	0.8964	0.8312	0.8312	0.7502	0.9483
QueryOnlyNugget docs	docs	query	na	0.7890	0.7890	0.5444	0.6668	0.6668	0.4600	0.8878



ragtime25										
nugget_coverage										
MAX_GRADE										
Judge	graded	nugget	preference	spearman	spearman@15	spearman@10	kendall	kendall@15	kendall@10	pearson
PrefNugget best	response	pairwise	best	0.8813	0.8813	0.8415	0.7594	0.7594	0.7047	0.9762
PrefNugget docs	docs	pairwise	best	0.8219	0.8219	0.7622	0.7223	0.7223	0.6593	0.9677
PrefNugget random	response	pairwise	random	0.8184	0.8184	0.7570	0.6986	0.6986	0.6286	0.9657
PrefNugget random docs	docs	pairwise	random	0.7717	0.7717	0.6951	0.6853	0.6853	0.6138	0.9672
GroundedNugget best	response	grounded	best	0.7552	0.7552	0.6728	0.6358	0.6358	0.5521	0.9780
GroundedNugget best docs	docs	grounded	best	0.7135	0.7135	0.6148	0.6617	0.6617	0.5804	0.9612
GroundedNugget random	response	grounded	random	0.6084	0.6084	0.4754	0.4958	0.4958	0.3771	0.9511
GroundedNugget random docs	docs	grounded	random	0.5675	0.5675	0.4220	0.4488	0.4488	0.3220	0.8872
QueryOnlyNugget	response	query	na	0.8155	0.8155	0.7538	0.6239	0.6239	0.5394	0.9687
QueryOnlyNugget docs	docs	query	na	0.8818	0.8818	0.8424	0.7818	0.7818	0.7333	0.9652

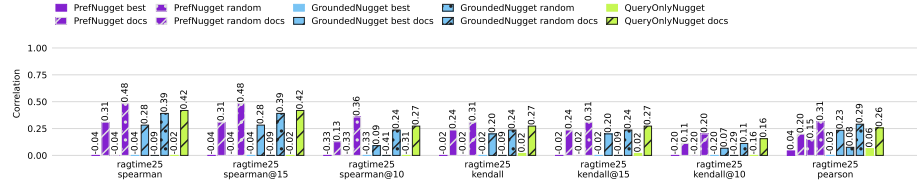


dragun25										
supportive										
MAX_GRADE										
Judge	graded	nugget	preference	spearman	spearman@15	spearman@10	kendall	kendall@15	kendall@10	pearson
PrefNugget best	response	pairwise	best	0.8855	0.6828	0.6121	0.7466	0.5168	0.4222	0.9184
PrefNugget docs	docs	pairwise	best	0.7079	0.7867	0.5061	0.5393	0.6312	0.3410	0.4021
PrefNugget random	response	pairwise	random	0.8764	0.4888	0.5350	0.7268	0.3479	0.3596	0.9276
PrefNugget random docs	docs	pairwise	random	0.7000	0.6486	0.5350	0.5235	0.5435	0.4166	0.3143
GroundedNugget best	response	grounded	best	0.8334	0.5869	0.1377	0.7009	0.4708	0.1054	0.8886
GroundedNugget best docs	docs	grounded	best	0.7056	0.6205	0.0519	0.5265	0.5287	0.0362	0.4017
GroundedNugget random	response	grounded	random	0.7641	0.5286	0.7246	0.6389	0.4128	0.5795	0.8442
GroundedNugget random docs	docs	grounded	random	0.6968	0.6089	0.2611	0.5141	0.5209	0.2236	0.2530
QueryOnlyNugget	response	query	na	0.7683	0.5464	0.6485	0.6047	0.3905	0.4667	0.8848
QueryOnlyNugget docs	docs	query	na	0.7676	0.6064	0.6079	0.5832	0.4712	0.4045	0.5286



2 Correlation with Truth Measure “Citation Support” (RQ4)

Judge	graded	nugget	preference	ragtime25						
				citation_support						
				MAX_GRADE						
				spearman	spearman@15	spearman@10	kendall	kendall@15	kendall@10	pearson
PrefNugget best	response	pairwise	best	-0.0685	-0.0685	-0.3781	-0.0556	-0.0556	-0.2501	0.1521
PrefNugget docs	docs	pairwise	best	0.3699	0.3699	0.2073	0.2037	0.2037	0.0682	0.2576
PrefNugget random	response	pairwise	random	0.3357	0.3357	0.1600	0.2077	0.2077	0.0698	0.2720
PrefNugget random docs	docs	pairwise	random	0.4703	0.4703	0.3415	0.2778	0.2778	0.1591	0.3135
GroundedNugget best	response	grounded	best	-0.1648	-0.1648	-0.5077	-0.1496	-0.1496	-0.3680	0.2233
GroundedNugget best docs	docs	grounded	best	0.5783	0.5783	0.4830	0.4282	0.4282	0.3386	0.3599
GroundedNugget random	response	grounded	random	0.0138	0.0138	-0.2717	0.0000	0.0000	-0.1886	0.3199
GroundedNugget random docs	docs	grounded	random	0.5904	0.5904	0.5016	0.4488	0.4488	0.3680	0.5168
QueryOnlyNugget	response	query	na	-0.0410	-0.0410	-0.3404	-0.0734	-0.0734	-0.2697	0.1429
QueryOnlyNugget docs	docs	query	na	0.4455	0.4455	0.3091	0.3455	0.3455	0.2444	0.3537

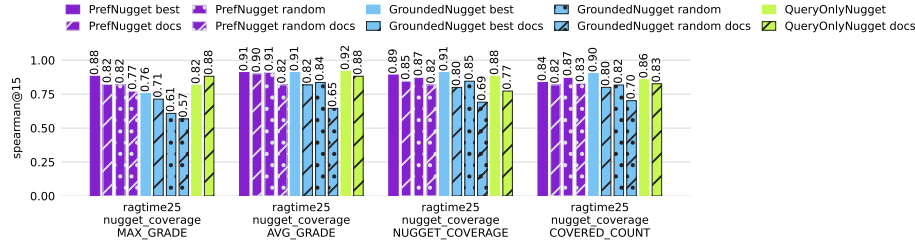


3 Various Nugget-Based Predicted Measures

Correlation=spearman@15

Judge	graded	nugget	preference	ragtime25			
				nugget_coverage			
				MAX_GRADE	AVG_GRADE	NUGGET_COVERAGE	COVERED_COUNT
PrefNugget best	response	pairwise	best	0.8813	0.9091	0.8909	0.8364
PrefNugget docs	docs	pairwise	best	0.8219	0.9000	0.8455	0.8182
PrefNugget random	response	pairwise	random	0.8184	0.9091	0.8727	0.8727
PrefNugget random docs	docs	pairwise	random	0.7717	0.8182	0.8182	0.8292
GroundedNugget best	response	grounded	best	0.7552	0.9091	0.9091	0.9021
GroundedNugget best docs	docs	grounded	best	0.7135	0.8182	0.8000	0.8000
GroundedNugget random	response	grounded	random	0.6084	0.8364	0.8455	0.8174
GroundedNugget random docs	docs	grounded	random	0.5675	0.6455	0.6909	0.7016
QueryOnlyNugget	response	query	na	0.8155	0.9182	0.8818	0.8565
QueryOnlyNugget docs	docs	query	na	0.8818	0.8818	0.7727	0.8265

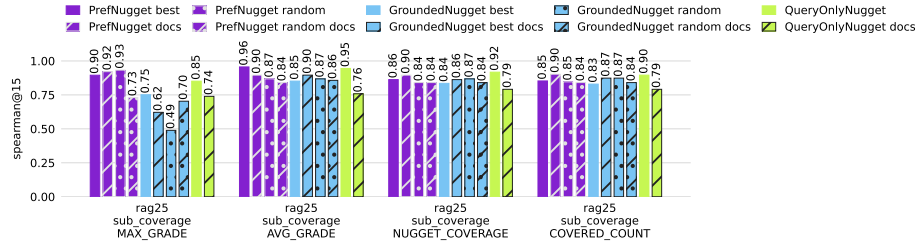
	Category	Comparison	Metric	N	median_diff	p-value	winner	sig
Category Wilcoxon Test	graded	docs vs response	spearman@15	20	-0.0506	0.0005	response	*
	nugget	grounded vs pairwise	spearman@15	16	-0.0773	0.0029	pairwise	*
	preference	best vs random	spearman@15	16	+0.0682	0.0018	best	*



Correlation=spearman@15

Judge	graded	nugget	preference	rag25			
				sub_coverage			
				MAX_GRADE	AVG_GRADE	NUGGET_COVERAGE	COVERED_COUNT
PrefNugget best	response	pairwise	best	0.8950	0.9560	0.8626	0.8528
PrefNugget docs	docs	pairwise	best	0.9212	0.8956	0.8956	0.9023
PrefNugget random	response	pairwise	random	0.9326	0.8681	0.8407	0.8462
PrefNugget random docs	docs	pairwise	random	0.7303	0.8407	0.8407	0.8407
GroundedNugget best	response	grounded	best	0.7511	0.8516	0.8352	0.8297
GroundedNugget best docs	docs	grounded	best	0.6216	0.8956	0.8626	0.8736
GroundedNugget random	response	grounded	random	0.4889	0.8681	0.8681	0.8736
GroundedNugget random docs	docs	grounded	random	0.7035	0.8571	0.8407	0.8407
QueryOnlyNugget	response	query	na	0.8512	0.9451	0.9176	0.8956
QueryOnlyNugget docs	docs	query	na	0.7393	0.7582	0.7912	0.7912

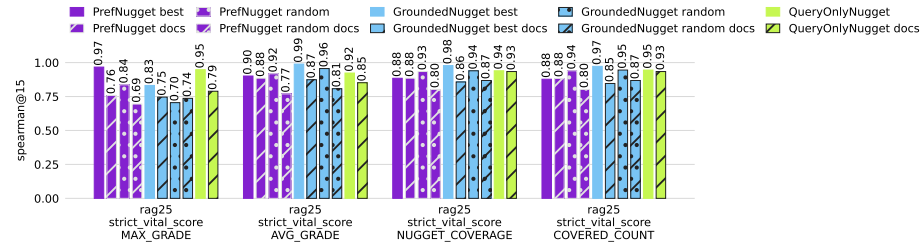
	Category	Comparison	Metric	N	median_diff	p-value	winner	sig
Category Wilcoxon Test	graded	docs vs response	spearman@15	20	-0.0192	0.2271	response	
	nugget	grounded vs pairwise	spearman@15	16	-0.0250	0.0309	pairwise	*
	preference	best vs random	spearman@15	16	+0.0275	0.0979	best	



Correlation=spearman@15

Judge	graded	nugget	preference	rag25			
				strict_vital_score			
				MAX_GRADE	AVG_GRADE	NUGGET_COVERAGE	COVERED_COUNT
PrefNugget best	response	pairwise	best	0.9669	0.9011	0.8846	0.8776
PrefNugget docs	docs	pairwise	best	0.7580	0.8846	0.8846	0.8831
PrefNugget random	response	pairwise	random	0.8363	0.9176	0.9341	0.9396
PrefNugget random docs	docs	pairwise	random	0.6943	0.7692	0.8022	0.8022
GroundedNugget best	response	grounded	best	0.8318	0.9890	0.9780	0.9725
GroundedNugget best docs	docs	grounded	best	0.7459	0.8736	0.8571	0.8462
GroundedNugget random	response	grounded	random	0.7050	0.9560	0.9396	0.9451
GroundedNugget random docs	docs	grounded	random	0.7366	0.8077	0.8081	0.8681
QueryOnlyNugget	response	query	na	0.9477	0.9231	0.9396	0.9451
QueryOnlyNugget docs	docs	query	na	0.7890	0.8516	0.9341	0.9341

	Category	Comparison	Metric	N	median_diff	p-value	winner	sig
Category Wilcoxon Test	graded	docs vs response	spearman@15	20	-0.1006	0.0004	response	*
	nugget	grounded vs pairwise	spearman@15	16	+0.0220	0.3257	grounded	
	preference	best vs random	spearman@15	16	+0.0357	0.0290	best	*



Correlation=spearman@15

Judge	graded	nugget	preference	dragun25			
				supportive			
				MAX_GRADE	AVG_GRADE	NUGGET_COVERAGE	COVERED_COUNT
PrefNugget best	response	pairwise	best	0.6828	0.5750	0.5832	0.5832
PrefNugget docs	docs	pairwise	best	0.7867	0.7929	0.7885	0.7885
PrefNugget random	response	pairwise	random	0.4888	0.5357	0.5857	0.5809
PrefNugget random docs	docs	pairwise	random	0.6486	0.6036	0.6143	0.6107
GroundedNugget best	response	grounded	best	0.5869	0.5107	0.5179	0.5179
GroundedNugget best docs	docs	grounded	best	0.6205	0.6750	0.7250	0.6750
GroundedNugget random	response	grounded	random	0.5286	0.6107	0.5964	0.6071
GroundedNugget random docs	docs	grounded	random	0.6089	0.6643	0.6786	0.6971
QueryOnlyNugget	response	query	na	0.5464	0.5571	0.2964	0.3354
QueryOnlyNugget docs	docs	query	na	0.6064	0.6643	0.4321	0.4884

	Category	Comparison	Metric	N	median_diff	p-value	winner	sig
Category Wilcoxon Test	graded	docs vs response	spearman@15	20	+0.1056	8.84e-05	docs	*
	nugget	grounded vs pairwise	spearman@15	16	-0.0516	0.1787	pairwise	
	preference	best vs random	spearman@15	16	+0.0255	0.1167	best	

